

AI Literacy

POD 2 - Introduction to AI

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My name is Prof. Suzanne Little, and I'm with the School of Computing here at Dublin City University, and I'll be one of the instructors on this course.

Before we look at the latest developments and how they might impact you, we should first consider what is meant by "artificial intelligence" and what technologies enable us to talk about "thinking machines".

One of the first examples of a thinking machine was a chess playing automaton that toured Europe in the 1700s and appeared to play a strong game of chess against human opponents. However, this was actually an illusion that allowed a human chess master hiding inside to operate the machine. Teaching computers to play games such as chess or go has been a key goal of programmers.

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Let's start with some important definitions. Firstly, we can consider a computer to be a machine that stores, retrieves and processes data or information. Computer programming is writing instructions to be executed by the computer. Computers now are incredibly common and significantly more powerful than the first machines, and the programmes they run are specific and complex.

To programme computers to perform tasks such as classification (apply a label to unseen data) or prediction (given data, what happens next), we use techniques from Machine Learning to identify patterns, rules or functions. Think of how you might have predictive text or spell-check in your text editor or on your phone. Machine learning also supports search engines that find a document



matching your query and recommender systems that suggest new music, movies or purchases based on your history.

So how is Artificial Intelligence different, and why are we talking about it so much now? Artificial Intelligence is computers or programmes that can “simulate human learning, comprehension, problem solving, decision making, creativity and autonomy” (IBM, 2024). In other words, AI has the ability to mimic a human or the ability to learn new skills that haven’t been explicitly programmed.

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We first started talking about AI in the 1950s when the idea was to use computers to model neural networks, that is functions that are meant to simulate how the human brain processes input. Mathematician Alan Turing proposed that a computer whose responses were indistinguishable from that of a human could be classed as “intelligent” - this is the Turing Test.

The first attempts at creating AI used a lot of handcrafted rules. In the mid-1960s there was a programme called ELIZA that was a chatbot therapist. It used pattern matching and sets of canned responses to pretend to interact. People took it quite seriously, but it was pretty clear that you were chatting with a machine.

In the 1970s & 1980s we had what is referred to as an “AI winter” where lots of promise and excitement for applications like machine translation, perception and speech understanding turned to disappointment and pessimism resulting in reductions in funding and research activity.

However, work continued and in 1997 Deep Blue from IBM became the first computer to defeat a reigning world chess champion under tournament conditions - a long way from our fake automaton! Then we started to move towards data-driven machine learning with the rise of the Internet and search companies like Google using these technologies to create more sophisticated tools using huge amounts of data.



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Progress was made in speech and video processing but the next big disruptive event was in the 2000s when the previous ideas of neural networks could finally be implemented thanks to increases in available computing power. Using Graphical Processing Units (GPUs), researchers were able to create many interconnected layers of functions (calculations) to imitate a neural network with huge improvements in applications like translation, computer vision and speech. The other ingredient was the availability of really large datasets from websites so more complex, data-rich machine learning tools became viewed as “AI”. This revolutionised applications like driver assistance systems, online translation, speech-to-text, ecommerce and scientific research.

Finally, we have the most recent twist in the story of AI with the public release of OpenAI’s ChatGPT, a large language model (LLM), in November 2022. This is the result of continuing increases in available computing power and training data and provides new functions like creating text or images. More than classification or prediction, these new models are Generative AI. So you can ask it to tell you a story, draft a speech or write new computer code and the output will be new and different. There are other LLMs and we’ll learn about them in the next pod.

So people use the term AI to refer to all sorts of data-driven machine learning through to the recent LLM chatbots and we will learn more about them in this course. For now, what AI are you using in your daily life? In the next activity you’ll identify all sorts of uses. And perhaps some you aren’t even aware of!